
Collaborative Real2Sim–Sim2Real Agential Learning for Geometric and Generative Human Dynamics

Weiqi Li · Zehao Zhang · Liang Lin · Guangrun Wang

Abstract Generating photorealistic, dynamic human–scene videos requires unifying geometric accuracy, motion expressiveness, and visual realism. While recent systems combine reconstruction, motion estimation, and diffusion-based generation, these modules often operate independently, leading to geometric drift and scene inconsistency. We introduce Collaborative Real2Sim–Sim2Real Agential Learning (CAL), a unified framework in which four specialized agents—Reconstruction, Motion Guidance, Consistency Critique, and Generation—exchange information through geometric and generative feedback. On the Real2Sim side, the Reconstruction Agent builds animatable human–scene Gaussian representations from monocular video, and the Motion Guidance Agent applies supplied poses and encodes rendered motion conditions. On the Sim2Real side, the Consistency Critique Agent assesses perceptual inconsistencies in rendered images and guides focused refinement, while the Generation Agent uses video diffusion to improve temporal appearance and human–scene integration. Localized critiques emphasize reconstruction losses in poorly reconstructed regions, and the Back-to-4D stage uses generated novel-view sequences as fixed pseudo-targets to refine the Gaussian model. Empirically, our system

outperforms the evaluated baselines on human video synthesis and reconstruction benchmarks, while demonstrating strong capabilities in reenactment and pose generalization. These results highlight the effectiveness of Real2Sim–Sim2Real agential learning in producing temporally stable, expressive, and visually coherent human–scene dynamics.

Keywords Pose-guided human video synthesis · 3D Gaussian splatting · Video diffusion · Multimodal critique · Collaborative agential learning

1 Introduction

Dynamic human modeling seeks to generate photorealistic videos of human subjects performing expressive, controllable motions while maintaining geometric accuracy and temporal consistency. Such capability enables personalized avatars and embodied telepresence, and offers a potential source of synthetic behavioral data for training embodied AI systems. Despite progress in pose-guided generation [7, 17, 32, 45], existing frameworks often treat reconstruction, motion estimation, and rendering as disjoint modules. This separation can lead to geometric drift, coarse detail preservation, and scene–human disharmony, especially under complex motion or lighting conditions.

This work seeks to unify geometric reasoning and generative synthesis under a single collaborative framework. We introduce four specialized agents: the **Reconstruction Agent** builds human–scene geometry from monocular videos using Gaussian-based deformation modeling; the **Motion Guidance Agent** applies supplied pose trajectories to the reconstructed human and encodes the resulting renderings as time-aware conditions; the **Consistency Critique Agent** identifies localized reconstruction errors and provides targeted

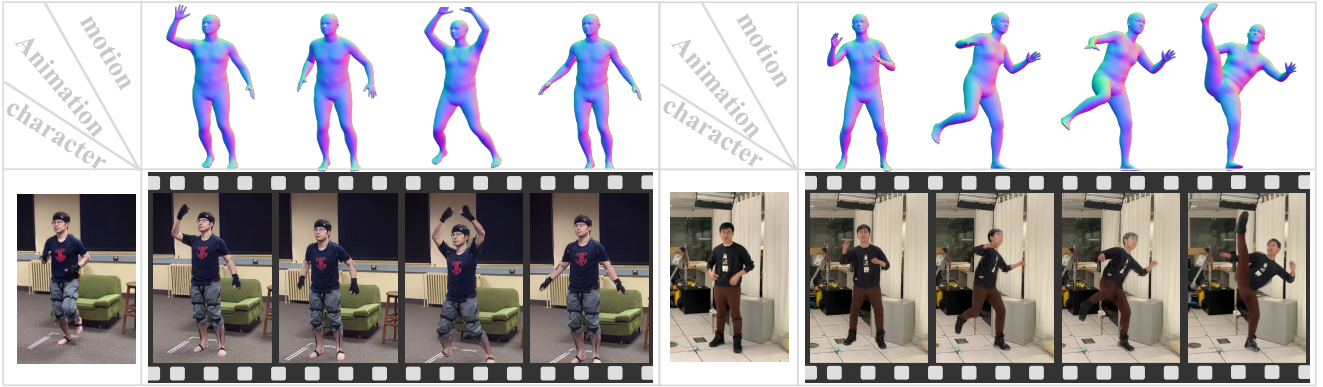
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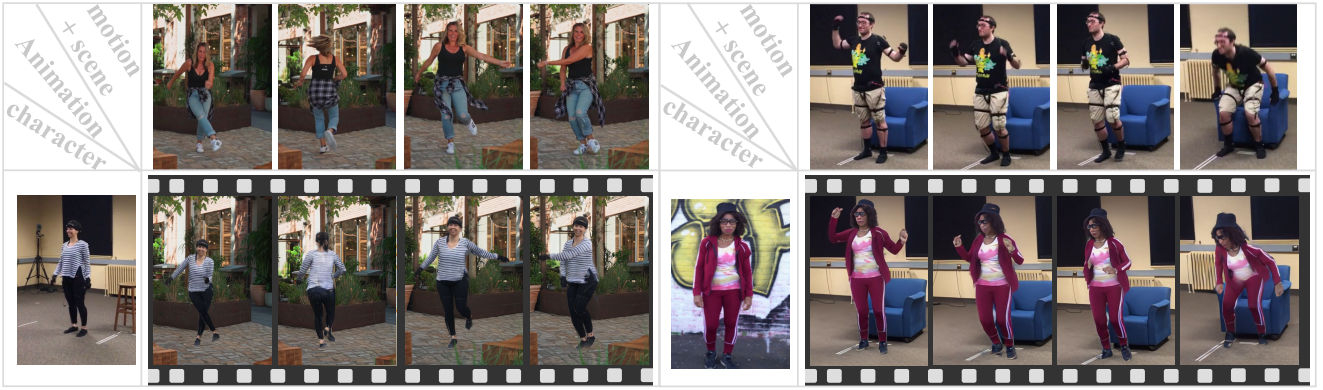
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(a) Animating characters with novel motions (e.g., text-generated motions)



(b) Video reenactment: teleporting characters into designated scenes guided by predefined motions

Fig. 1 Compelling applications of our proposed method in human dynamics synthesis highlight four key strengths: temporal consistency, geometric plausibility, expressive motion handling, and seamless integration into target scenes. These capabilities enable a wide range of applications, including but not limited to: (a) animating characters with novel motions—either drawn from motion capture datasets such as AMASS [22] or synthesized from textual descriptions [6]; and (b) video reenactment, where characters are smoothly inserted into target scenes and animated according to predefined motion trajectories.

refinement signals through multimodal reflection; and the **Generation Agent** performs diffusion-based video synthesis, with refined novel views providing feedback for subsequent reconstruction. Together, these agents form a geometric–generative loop linking body structure, dynamic motion, and photorealistic rendering (Figure 2). Here, each agent denotes a specialized module with a concrete role in the pipeline.

The central challenge is to synthesize human–scene videos from monocular inputs while preserving geometry, identity, motion, and visual realism beyond the observed frames. This involves four related problems. *Geometric consistency* requires stable body structure and identity under large articulation, occlusion, and view-point changes. *Fine-grained reconstruction* requires resolving blurred limbs, hands, and unclear separation between the feet and the floor, even when these errors occupy only small image regions. *Motion and view-point generalization* requires handling unseen poses and incomplete camera coverage. Finally, *human–scene integration* requires coherent appearance, lighting, and

temporal detail when the reconstructed human moves within the original or a target scene.

Explicit human reconstruction provides a structural basis for these tasks. SMPL [20] supplies parametric articulation, while deformable Gaussian models learn subject-specific geometry and appearance [10, 15, 16, 19, 27]. Video diffusion complements this structural control with learned visual detail and temporal appearance modeling [7, 17, 45]. Their combination motivates a framework in which reconstruction guides generation and visual feedback directs further refinement of the reconstructed model.

To address these challenges, we introduce **Collaborative Real2Sim–Sim2Real Agential Learning (CAL)** to link geometric modeling and generative synthesis through iterative feedback. On the **Real2Sim** side, the Reconstruction Agent recovers an animatable digital human and scene from monocular observations, and the Motion Guidance Agent applies supplied pose trajectories and encodes the resulting renderings as motion conditions. On the **Sim2Real** side, the Con-

sistency Critique Agent assesses rendered results for perceptual inconsistencies, and the Generation Agent synthesizes temporally coherent, photorealistic videos. These terms describe the agents’ complementary functional roles. “Sim” refers to the animatable digital scene used for rendering and articulation; the Sim2Real side targets perceptual consistency and visual realism.

The two feedback mechanisms connect the modules through concrete training operations. Critique-to-geometry refinement converts region-level feedback into localized reconstruction-loss weighting, focusing additional optimization on poorly reconstructed areas. Back-to-4D refinement renders new camera views, enhances their appearance with the Generation Agent, and uses the resulting sequences as fixed pseudo-targets for Gaussian updates. These operations allow generation to inform reconstruction and critique to guide optimization while maintaining the geometric model as the basis for motion and viewpoint control.

Figure 1 illustrates applications to novel-motion animation and target-scene reenactment. Novel motions are drawn from AMASS [22] or synthesized from textual descriptions using MoMask [6]. CAL applies these externally supplied motions to the personalized avatar and generates the corresponding video. For target-scene reenactment, the human Gaussian model is placed in the designated scene and animated with a predefined trajectory before video refinement. These applications demonstrate expressive motion handling and human-scene integration alongside the quantitative reconstruction and synthesis evaluations.

Our contributions are four-fold:

- **Collaborative geometric-generative framework.** We connect reconstruction, motion guidance, critique, and generation through a Real2Sim-Sim2Real workflow in which geometric renderings condition video synthesis and visual feedback supports reconstruction refinement.
- **Learnable motion transport.** We extend standard LBS with learned residual skinning weights and non-rigid Gaussian offsets, enabling subject-specific deformation under diverse poses.
- **Critique-to-geometry mechanism.** We convert non-differentiable multimodal textual and spatial feedback into localized reconstruction-loss reweighting, connecting semantic critique with gradient-based geometric optimization.
- **Back-to-4D generative distillation.** We transfer diffusion-based generative priors into 3D Gaussian attributes through fixed novel-view pseudo-targets, enriching appearance in unobserved regions while retaining pose-driven control.

Rather than treating geometric and generative modeling as isolated modules, CAL integrates them into a feedback process for geometrically controlled, visually coherent human-scene dynamics. Supplementary videos and project materials are available at <https://liwq229.github.io/cal/>.

2 Related Work

2.1 Human-Scene Reconstruction

Reconstructing humans and scenes from monocular video has seen substantial progress through advances in neural implicit representations and radiance fields. Methods such as [10, 15, 16, 19, 25, 27] use NeRF-based models [24] or 3D Gaussian Splatting (3DGS) [13, 28] to represent deformable human bodies in 3D. Joint human-scene approaches such as NeuMan [12] also model the person and the surrounding environment. Recovering fine local details and maintaining coherent appearance under novel poses and views remain challenging. CAL couples Gaussian-based human-scene reconstruction with localized visual critique. The **Reconstruction Agent** learns subject-specific geometry and rigid and non-rigid deformation, while the **Consistency Critique Agent** identifies poorly reconstructed regions and guides iterative refinement through regional reconstruction-loss weighting.

2.2 Human Video Generation

Pose-guided human video generation aims to animate a target identity according to user-specified motion inputs. Recent methods [17, 21, 34, 42] generate high-quality sequences conditioned on 2D keypoints or pose sequences. Broader controllable video methods use motion and image instructions [36, 40]. For example, AnimateAnyone [17] introduces ReferenceNet to preserve identity and applies a pose guider for motion control. Champ [45] uses 3D parametric guidance for shape alignment and motion control. Human4DiT [32] models correlations across viewpoints and time, while MIMO [23] uses spatial decomposition to control the character, motion, and scene. Maintaining identity and structural consistency under complex motions and viewpoint changes remains a central challenge. CAL introduces a **Motion Guidance Agent** that applies supplied poses to the personalized Gaussian avatar and encodes the resulting renderings as time-aware motion conditions, and a **Generation Agent** that uses video diffusion to improve temporal appearance and human-scene integration. Refined novel-view sequences also

serve as fixed pseudo-targets for further reconstruction updates, connecting video generation with the animatable geometric model.

2.3 Real2Sim and Sim2Real

Sim-to-Real (Sim2Real) methods aim to transfer models trained in simulation to the real world, typically through domain randomization, domain adaptation, or improved physical modeling. Conversely, Real-to-Sim (Real2Sim) focuses on incorporating real-world observations—geometry, appearance, or dynamics—into simulation to reduce the sim-real gap and construct more faithful digital environments.

RoboGSim [18] and DISCOVERSE [11] combine reconstruction of real environments with high-fidelity simulation to support robot learning and Sim2Real transfer. In perceptual modeling, neural rendering and Gaussian-based representations enable photorealistic, geometry-aware reconstruction. Our work adopts this bidirectional perspective for geometric-generative human dynamics. The Real2Sim side comprises the **Reconstruction Agent** and **Motion Guidance Agent**, which build an animatable human-scene model and encode rendered motion conditions. The Sim2Real side comprises the **Consistency Critique Agent** and **Generation Agent**, which assess visual inconsistencies and refine rendered videos. Localized critique and generated novel-view pseudo-targets provide feedback for subsequent reconstruction updates.

3 Method

3.1 Task Definition and Framework Overview

The input is a monocular video $I_{1:N}$ of one person in a scene. Reconstruction uses camera information, foreground masks, and estimated SMPL parameters associated with the selected observations. For animation, the user supplies a target pose sequence $\theta_{1:T}$ and camera sequence $K_{1:T}$; a target scene can optionally replace the original static background. The outputs are an animatable human Gaussian model H_G , a static-scene model S_G , and a synthesized video $\hat{I}_{1:T}$. The target motion can reproduce the captured motion or come from an external source. The target camera can likewise follow observed or new viewpoints.

Table 1 summarizes the notation for geometry, rendered conditions, and generated images. Table 2 summarizes the supervision and trainable components of each stage.

Table 1 Objects used by the method.

Symbol	Meaning
I_t	Observed RGB frame
H_G, S_G	Human and static-scene Gaussians
β, θ_t	Subject shape and supplied body pose
K_t	Camera for a rendered frame
C_t	Rendered image condition
z_t	Encoded image latent
$h_t^{(l)}$	Condition feature at level l
(b_j, f_j)	Critic box and feedback label
$\hat{I}_t$	Fixed generated pseudo-target

Figure 2 groups Reconstruction and Motion Guidance on the Real2Sim side, and Consistency Critique and Generation on the Sim2Real side. Reconstruction produces the human-scene model that Motion Guidance renders and encodes for Generation. Critique guides localized reconstruction updates, while Back-to-4D uses generated novel views as fixed targets for further Gaussian refinement.

3.2 Reconstruction Agent: Human-Scene Gaussian Modeling

3.2.1 Initialization and Subject Factorization

We represent the human and static environment separately with 3D Gaussians [13]. A Gaussian stores a center, scale, rotation, opacity, and spherical-harmonic appearance coefficients. These quantities support differentiable projection and alpha compositing. The scene Gaussians are initialized from structure-from-motion using COLMAP [30, 31]. The human Gaussians are initialized from a subject-specific SMPL mesh [20] in a canonical Da pose.

The canonical human model provides a common location for subject appearance across the captured frames. The implementation samples 50K points from the SMPL mesh and initializes their skinning weights from SMPL. Gaussian adaptive control follows GauHuman [10]. Subject details such as hair and clothing are learned through the Gaussian attributes and deformation model rather than being restricted to the bare body surface.

3.2.2 Learnable Motion Transport

The Reconstruction Agent learns a motion-transport model that combines canonical non-rigid offsets with residual-corrected skinning. A spatial feature encoder provides a feature f_k^c at canonical Gaussian k . The non-rigid decoder $\mathcal{D}_{nr}$ predicts offsets for its position, scale,

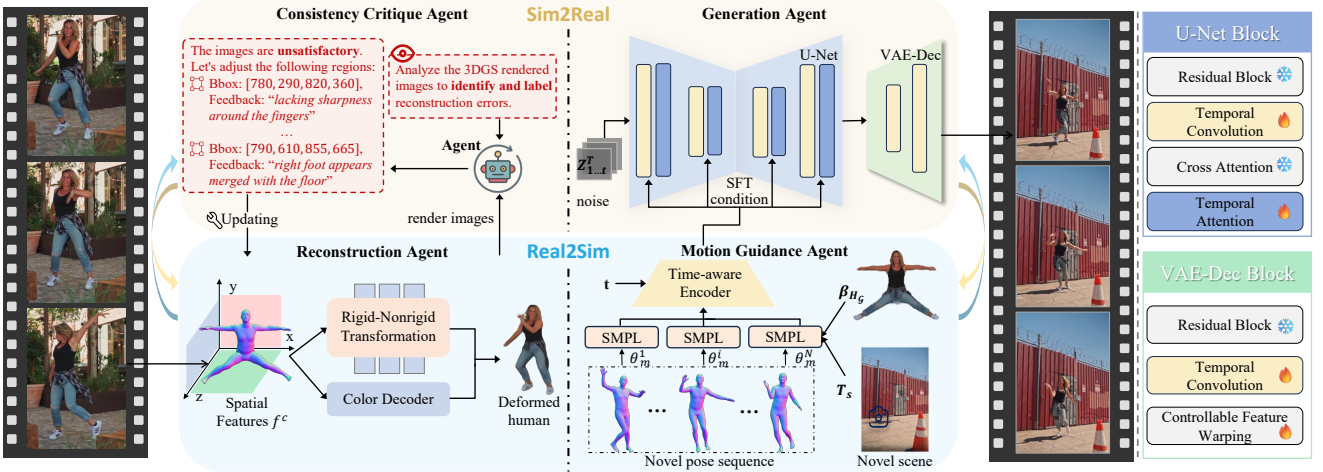


Fig. 2 Illustration of the Collaborative Agential Learning. The lower **Real2Sim** group comprises the **Reconstruction Agent** and **Motion Guidance Agent**: Reconstruction Agent recovers 3D human–scene structures from monocular video by modeling rigid and non-rigid deformations, while Motion Guidance Agent applies supplied parametric pose sequences and encodes the resulting human renderings into time-aware motion conditions. The upper **Sim2Real** group comprises the **Consistency Critique Agent** and **Generation Agent**: Consistency Critique Agent identifies visually inconsistent regions and provides feedback for focused reconstruction updates, while Generation Agent employs Spatial Feature Transform (SFT) [35] inside a video-diffusion rendering pipeline to synthesize temporally coherent, photorealistic outputs. The two groups are coupled through rendering-based conditioning, localized critique feedback, and Back-to-4D refinement.

Table 2 Stages, supervision, and parameter scope. “Fixed” refers to the update in the corresponding row.

Stage	Input and output	Supervision	Updated or fixed components
Initial reconstruction	Selected observations to human and scene Gaussians	Observed colors, masks, structural and perceptual image losses	Gaussian attributes and reconstruction decoders are trained
Local critique	Current renderings to boxes and feedback, then refined geometry	Original images, with additional emphasis on flagged regions	Reconstruction is updated; MLLM evaluation is fixed
Video-model training	Rendered conditions and training clips to predicted video	Ground-truth video; sequence decoding losses	Temporal layers and CFW are trained; pretrained spatial layers are fixed as described below
Pose-driven synthesis	Supplied motion, camera, and scene to rendered conditions and video	Target poses are external controls	Trained networks and avatar are used for inference
Back-to-4D	New-camera renderings to refined pseudo-targets and updated Gaussians	Fixed generated images	Gaussian model is optimized; generated targets and video model are fixed

and rotation:

$$(\Delta \mathbf{x}_k, \Delta \mathbf{s}_k, \Delta \mathbf{q}_k) = \mathcal{D}_{nr}(f_k^c). \quad (1)$$

The deformed canonical position and positive scale are

$$\mathbf{x}_k^d = \mathbf{x}_k^c + \Delta \mathbf{x}_k, \quad (2)$$

$$\mathbf{s}_k^d = \mathbf{s}_k^c \odot \exp(\Delta \mathbf{s}_k). \quad (3)$$

Rotation uses the recorded quaternion residual composition, detailed in Appendix A. These offsets allow the Gaussian model to adjust to appearance and deformation beyond the underlying skeleton.

A separate decoder $\mathcal{D}_r$ predicts residuals to the initialized LBS weights. Let w_{ki}^c be the initialized influence

of joint i on Gaussian k , and let δw_{ki} be the learned residual. A supplied pose determines joint transforms $B_i(\beta, \theta_t)$. In homogeneous coordinates, the articulated center is

$$\bar{\mathbf{x}}_{k,t}^p = \sum_{i=1}^{24} (w_{ki}^c + \delta w_{ki}) B_i(\beta, \theta_t) \bar{\mathbf{x}}_k^d. \quad (4)$$

The learned non-rigid offsets and corrected LBS weights define the subject-specific articulation applied by the Motion Guidance Agent.

The construction keeps identity-dependent shape β associated with the subject, while θ_t varies over the requested motion. After optimization, skinning weights and Gaussian attributes are stored for direct animation.

This cached representation avoids evaluating the spatial feature encoder and reconstruction decoders for every animation frame in the reported implementation.

3.2.3 Rendering and Scene Composition

For a camera K_t , the rendering condition is

$$C_t = \text{Render}(T_s(H_G(\beta, \theta_t)) \cup S_G; K_t). \quad (5)$$

Here T_s places the articulated human in the scene, and the union indicates compositing the corresponding Gaussian sets in the renderer. Appendix A.1 recalls the Gaussian covariance and alpha-compositing equations.

For target-scene animation, we replace the source scene Gaussians with a reconstructed target scene and render the articulated human at the user-specified placement and pose. These renderings condition the video generator.

3.2.4 Reconstruction Supervision

Reconstruction uses color, mask, structural, and perceptual terms:

$$\mathcal{L}_g = \mathcal{L}_{\text{color}} + \lambda_m \mathcal{L}_{\text{mask}} + \lambda_s \mathcal{L}_{\text{struct}} + \lambda_p \mathcal{L}_{\text{LPIPS}}. \quad (6)$$

The coefficients are $\lambda_m = 0.5$ and $\lambda_s = \lambda_p = 0.01$. Color uses an ℓ_1 discrepancy between the observed and rendered images. The mask term compares the rendered foreground with the reference segmentation using an ℓ_2 discrepancy. The structural term $\mathcal{L}_{\text{struct}}$ is an SSIM-based reconstruction discrepancy, and the perceptual term uses LPIPS [41].

3.3 Motion Guidance Agent: Pose-Driven Rendering and Conditioning

For replay of captured motion, poses are obtained from a pretrained SMPL estimator [5]. Novel poses can come from AMASS motion-capture sequences or the text-to-motion model MoMask.

The Motion Guidance Agent applies the learned motion transport in Equation (4) to the supplied poses and renders the resulting sequence under Equation (5). These renderings encode the subject’s silhouette, clothing, and body proportions alongside the target pose.

The VAE image encoder maps rendered images to latent inputs. A time-aware condition encoder then extracts sequence-sensitive features:

$$z_{1:T} = E_{\text{VAE}}(C_{1:T}), \quad h_{1:T}^{(l)} = E_{\text{cond}}^{(l)}(z_{1:T}). \quad (7)$$

The index l denotes a level of the condition encoder’s feature hierarchy. The conditioning path uses a compact

encoder based on the contracting path of the diffusion U-Net, with temporal processing. It captures the visual changes caused by the given motion.

The resulting features condition the Generation Agent through SFT layers (Section 3.5).

3.4 Consistency Critique Agent: Localized Feedback and Refinement

3.4.1 Input Screening and Region Feedback

The Consistency Critique Agent assesses reconstruction images with a multimodal language model. Before reconstruction, a screening prompt compares the source person with an SMPL-based rendering and returns a score and analysis to identify inaccurate body estimates (Appendix C).

The first reflection round renders a set of images from the training viewpoints using the initial reconstruction. Leveraging prior knowledge of common 3DGS failures, such as blurriness, unnatural body-part intersections, and ground-fusion artifacts, the MLLM localizes poorly reconstructed regions and outputs a feedback label for each:

$$\mathcal{B}^{(r)} = \{(b_j^{(r)}, f_j^{(r)})\}_j = Q(\mathcal{I}^{(r)}, \text{prompt}). \quad (8)$$

Here r indexes the reflection round, $\mathcal{I}^{(r)}$ is the set of rendered images, $b_j^{(r)}$ specifies an identified region by its 2D bounding box, and $f_j^{(r)}$ is the corresponding textual feedback label. Figure 3 shows the interaction between the Reconstruction Agent and the Consistency Critique Agent. Figure 4 illustrates the resulting refinement of fine detail and feet–floor separation.

We use pretrained Qwen2.5-VL-32B [1, 29] as the perceptual critic and keep its parameters fixed.

3.4.2 Critique-to-Geometry Mechanism

This mechanism converts the critic’s spatial boxes and textual feedback into regional weighting of reconstruction losses. Let $\Omega(b_j)$ denote the image region inside a box flagged by negative feedback. Additional reconstruction training emphasizes those regions while preserving supervision from the original frames. A compact description is

$$\mathcal{L}_{\text{local}} = \sum_j \ell_{\Omega(b_j)}(\hat{C}, C, \hat{M}, M), \quad \mathcal{L}_c = \mathcal{L}_g + \omega \mathcal{L}_{\text{local}}. \quad (9)$$

The regional loss ℓ_{Ω} uses the color, mask, structural, and perceptual reconstruction terms over the flagged regions, with additional weight ω .

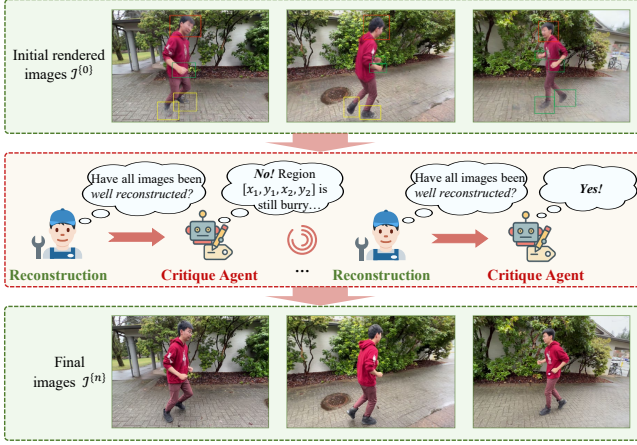


Fig. 3 Overview of Interaction between the Reconstruction Agent and the Consistency Critique Agent. We introduce a generic AI agent system for reconstruction via MLLM models.

The critic determines the regions and labels. The renderer computes the current predicted image, and the reconstruction loss supplies gradients to the trainable Gaussian and reconstruction parameters. No gradient is propagated through the language-model response or its text.

3.4.3 Finite Reflection and Selection

If all feedback labels indicate that the regions are “well reconstructed,” the process concludes and the reconstruction from that round is selected. If any regions receive negative feedback, a subsequent training round focuses on those localized areas. To avoid an infinite loop, we set a maximum number of rounds. If this limit is reached, we select the reconstruction with the smallest set of poorly reconstructed regions.

3.5 Generation Agent: Temporally Consistent Video Synthesis

3.5.1 Rendering-Conditioned Denoising

The Generation Agent processes rendered conditions with a temporal video diffusion U-Net. Its initialization comes from Stable Diffusion V2.1. Temporal convolution and attention augment spatial processing. The model is trained on short sequences so that appearance is processed with temporal context rather than as independent images. Table 14 specifies temporal mixing in the middle block and details the spatial stages.

The Generation Agent receives the condition features $h_{1:T}^{(l)}$ from the Motion Guidance Agent and injects them into its denoising U-Net through Spatial Feature

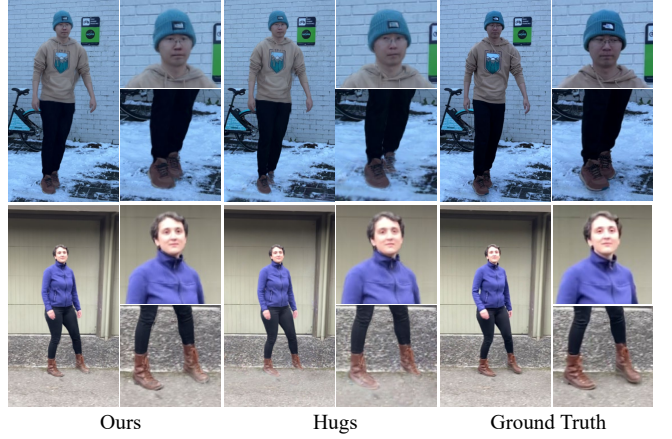


Fig. 4 Visualization of Consistency Critique Agent’s results. Our method effectively eliminates artifacts such as unclear separation between the feet and the floor.

Transform (SFT) [35] layers. For an intermediate feature $F_t^{(l)}$, learned mappings from the condition predict affine modulation parameters $a_t^{(l)}$ and $b_t^{(l)}$:

$$(a_t^{(l)}, b_t^{(l)}) = \Psi^{(l)}(h_t^{(l)}), \quad (10)$$

$$\text{SFT}(F_t^{(l)}) = a_t^{(l)} \odot F_t^{(l)} + b_t^{(l)}. \quad (11)$$

The feature-level multiplication and addition provide spatially varying conditioning. This conditioning encourages adherence to the rendered pose while allowing appearance refinement.

Latent Space Fine-Tuning (LSFT) updates the U-Net’s temporal layers while keeping its pretrained spatial layers fixed [2, 44]. At sampling time, the method uses progressive patch aggregation [33] for arbitrary output resolutions and motion-guided diffusion sampling [39] to connect frame information through optical-flow guidance.

3.5.2 Temporal Decoding and Feature Warping

Even a temporally processed latent sequence can lose consistency during image decoding. The Generation Agent therefore uses temporal residual processing in the VAE decoder. The training combines image fidelity, perceptual similarity, frame-difference consistency, and a patchwise temporal adversarial term. For generated frames $\hat{I}_{1:T}$ and reference frames $I_{1:T}^{\text{gt}}$, the decoder objective is

$$\mathcal{L}_{\text{dec}} = \mathcal{L}_1 + \mathcal{L}_{\text{LPIPS}} + \alpha \mathcal{L}_{\text{diff}} + \beta_{\text{adv}} \mathcal{L}_{\text{GAN}}, \quad (12)$$

Algorithm 1 Back-to-4D Generative Distillation

Require: Initial human-scene Gaussians $\mathcal{G}$; trained video model D ; supplied poses; novel cameras; configured refinement budget

- 1: **for** each configured refinement pass **do**
- 2: Render a pose-driven sequence $C_{1:T}$ from $\mathcal{G}$ at the novel cameras.
- 3: Refine the sequence with D to obtain $\tilde{I}_{1:T}$.
- 4: Hold $\tilde{I}_{1:T}$ and D fixed.
- 5: Optimize Gaussian parameters using the rendered-image reconstruction losses against the fixed pseudo-targets.
- 6: **end for**
- 7: **return** Refined animatable Gaussian model

where $\alpha = 0.5$ and $\beta_{\text{adv}} = 0.025$. The frame-difference term compares temporal changes:

$$\mathcal{L}_{\text{diff}} = \sum_{t=1}^{T-1} \left\| (\hat{I}_{t+1} - \hat{I}_t) - (I_{t+1}^{\text{gt}} - I_t^{\text{gt}}) \right\|_1. \quad (13)$$

It encourages agreement with observed interframe differences, rather than simply penalizing all motion. The adversarial term uses a temporal patch discriminator, and LPIPS provides perceptual image comparison [41].

Controllable Feature Warping (CFW) transfers encoder information to support decoding [33, 43]. Training freezes pretrained spatial layers and updates temporal convolution and CFW components. The decoder is trained with generated latent sequences from the fine-tuned U-Net and corresponding ground-truth sequences. Training configurations and sampling settings are provided in Appendix A.

3.5.3 Back-to-4D Generative Distillation

Constructing an articulated human model from a video often suffers from incomplete input coverage, leading to gaps in reconstruction. Inspired by image-guided reconstruction refinement [8], we implement Back-to-4D generative distillation through Iterative Latent Projection (ILP).

In each iteration, we first generate synthetic views with novel camera poses or trajectories from the reconstructed human-scene Gaussians. These views are then denoised using the Generation Agent. The enhanced outputs serve as fixed pseudo-targets to refine and update the 3DGS. This refinement takes approximately 2.5k training iterations. Algorithm 1 summarizes the procedure.

Gaussian updates optimize position, appearance, opacity, scale, and rotation through the differentiable renderer.

4 Experiments

4.1 Data, Implementation, and Evaluation Scope

4.1.1 Data Sources

Everyday videos frequently include camera motion, which facilitates our 3D human reconstruction process. We curated an in-the-wild monocular video dataset from online repositories. The MoVi dataset [4] provides 225 videos with distinct individuals performing full-body motions in indoor settings. Additionally, we collected 1K single-person videos from HumanVid [38], covering a range of ages and capturing various perspectives, including full-body, half-body, and close-up shots, in both indoor and outdoor environments.

NeuMan [12] provides the human-scene reconstruction benchmark and the testbed for the Consistency Critique Agent. We report video synthesis comparisons on HumanVid and Generation Agent ablations on MoVi. The temporal decoder is trained first on MoVi and then on HumanVid, following the schedules in Appendix A. To assess pose generalization and animation capabilities, we further drive CAL with novel motion trajectories from AMASS [22] and MoMask [6]. These sequences provide motion signals, while visual identity comes from the subject-specific reconstruction.

4.1.2 Implementation

We use Qwen2.5-VL-32B [1, 29] for critique and initialize the Generation Agent from Stable Diffusion V2.1. Network architectures, training configurations, and sampling settings are detailed in Appendix A.

4.1.3 Synthesis Comparison Protocol

We compare our method with prior state-of-the-art approaches, namely AnimateAnyone [17], AniGS [28], MIMO [23], and Champ [45]. We use the third-party implementation of AnimateAnyone at <https://github.com/MooreThreads/Moore-AnimateAnyone>, while the official implementations are used for the other methods.

These approaches use a single input reference image. We split the monocular video into multiple subsequences with a stride of 10. Our Reconstruction Agent takes all first frames of the subsequences as its training set, while the other methods use the first frame of each subsequence as the reference image for that subsequence. All methods are then evaluated on the test frames. Thus CAL uses a collection of reference frames to prepare a subject model, whereas each baseline synthesis call uses one reference image.

Table 3 Quantitative comparisons with video synthesis models on HumanVid [38] dataset.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID-VID $\downarrow$	User $\uparrow$
AnimateAnyone	16.102	0.633	0.398	99.49	2.482
AniGS	18.974	0.649	0.373	86.62	3.136
MIMO	20.845	0.783	0.266	82.34	3.504
Champ	17.986	0.651	0.382	101.91	2.637
Ours	21.934	0.802	0.223	76.20	4.103

4.2 Metrics and User Evaluation

We assess image quality using PSNR, SSIM [37], and LPIPS [41]. Higher PSNR and SSIM and lower LPIPS indicate better reconstruction or synthesis quality. Video comparisons additionally use the FID-VID score reported in the experiments [9], with lower values indicating better quality.

To quantitatively evaluate the temporal stability of generated videos, we employ a flow-based temporal consistency metric. This metric measures the pixel-wise discrepancy between consecutive frames, accounting for motion via optical flow. Let $F_{t \rightarrow t+1}$ be the estimated flow, $M_t(p)$ an occlusion mask that selects visible pixels, and N_t the number of valid pixels. The temporal error is

$$e_t(p) = I_t(p) - \mathcal{W}(I_{t+1}, F_{t \rightarrow t+1})(p), \quad (14)$$

$$E_{\text{temp}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{N_t} \sum_p \|M_t(p)e_t(p)\|_2^2. \quad (15)$$

Here $\mathcal{W}$ warps I_{t+1} to the coordinate system of I_t using the optical flow. For the Temporal Consistency Score (T-Cons) in our ablation studies, we normalize the error to a $[0, 1]$ scale:

$$\text{T-Cons} = \exp(-\gamma E_{\text{temp}}), \quad \gamma = 1.0. \quad (16)$$

A higher T-Cons indicates greater temporal stability under this flow-based evaluation. We report T-Cons for the module ablations in Table 11.

We also conduct a user study to evaluate the visual quality of the generated videos. We randomly select 15 samples and collect anonymous ratings from 25 participants, with the highest score being 5. The final User score is obtained by averaging the ratings. The questionnaire evaluates motion fidelity, identity preservation, and visual realism; the full protocol is provided in Appendix B.1.

4.3 HumanVid Synthesis Comparison

Quantitative comparison. As shown in Table 3, for complex and large-scale motions with camera movement,

the proposed method achieves higher PSNR and SSIM values while reducing LPIPS and FID-VID scores compared with the evaluated methods. CAL also obtains the highest User score in Table 3, consistent with the quantitative evaluation and supporting the perceptual quality of the generated videos.

Qualitative comparison. Figure 5 illustrates the results of different pose-driven video generation methods applied to complex or large-scale human motions. AniGS reconstructs a consistent human avatar using 4DGS but does not account for scene illumination, resulting in unrealistic visuals in these examples. Champ incorporates multiple 2D motion signals projected by a shape-aligned SMPL model, but still suffers from artifacts, particularly in facial rendering. While MIMO also decomposes the 2D video in a 3D-aware manner, its multi-faceted encoding mechanisms may compromise the stability of diffusion-based generation when the character performs large-angle rotations, as illustrated by the facial inconsistencies in the first and third rows of Figure 5.

In contrast, the Reconstruction Agent provides a reliable 3D reference for consistent appearance. When handling large-scale motions, the Motion Guidance Agent aligns the supplied pose with the personalized avatar, maintaining consistency during complex movements. Additional motion and scene applications are presented in Section 4.9.

4.4 NeuMan Reconstruction Comparison

Table 4 compares our spatial fine-grained reconstruction method with Vid2Avatar, NeuMan, and HUGS on the NeuMan dataset [12]. Our method achieves the best SSIM, PSNR, and LPIPS among these methods. Figure 4 shows the corresponding improvement in visual detail, including clearer separation between the feet and the floor. These results demonstrate the effectiveness of the Consistency Critique Agent in identifying and refining local reconstruction artifacts.

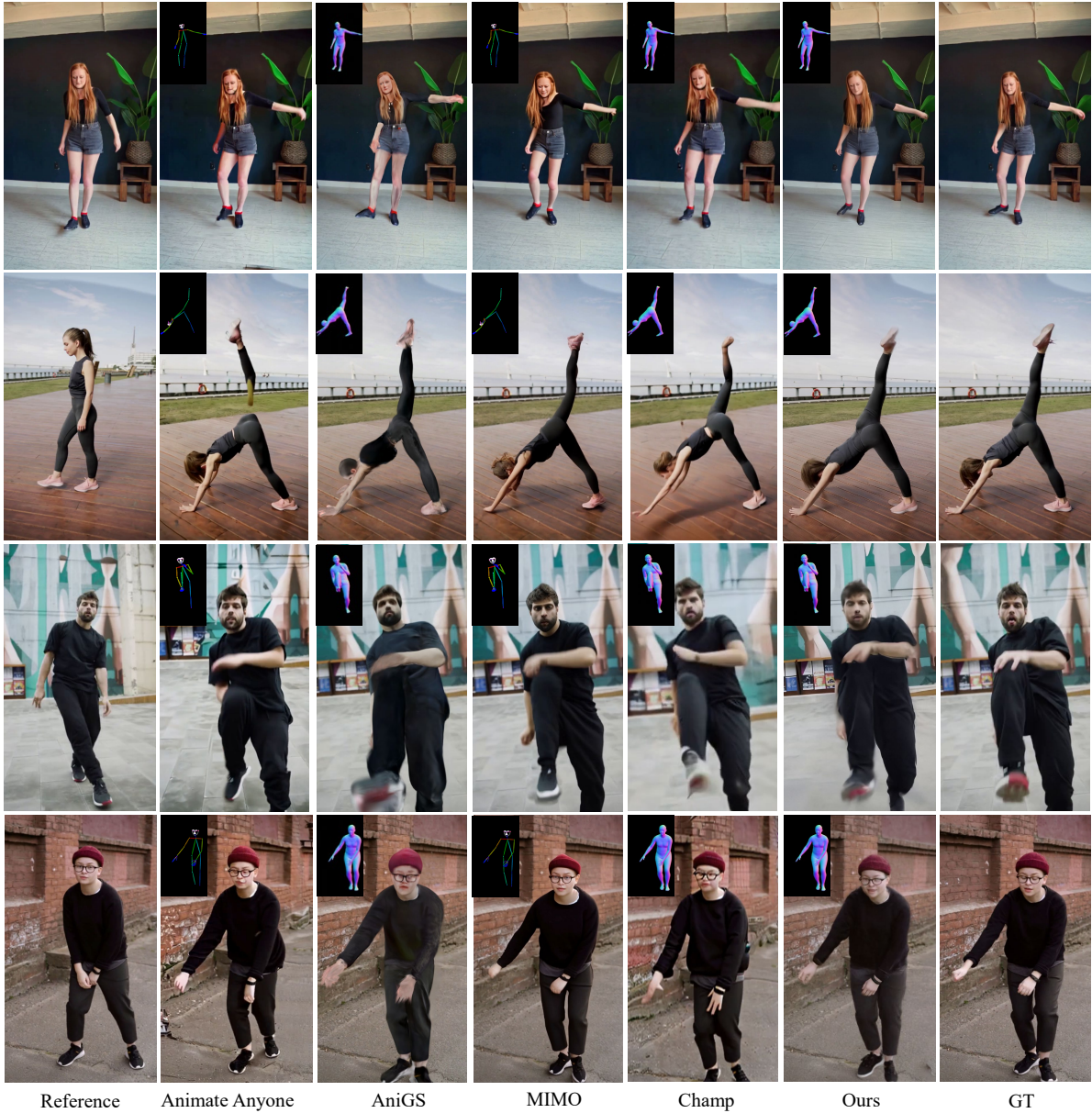


Fig. 5 Qualitative comparisons with state-of-the-art methods on the HumanVid dataset demonstrate that our approach consistently delivers high-quality results, even in the presence of large motions. Please zoom in to better observe the detailed differences.

Table 4 Quantitative comparisons with 3D reconstruction methods on NeuMan [12] dataset.

Method	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$
Vid2Avatar	0.566	16.192	0.637
NeuMan	0.793	24.543	0.265
HUGS	0.829	25.498	0.122
Ours	0.891	26.992	0.081

4.4.1 Per-Scene NeuMan Reconstruction

Table 5 compares reconstruction quality on six NeuMan scenes using PSNR, SSIM, and LPIPS. CAL achieves

the highest PSNR and lowest LPIPS in all six scenes, and the highest SSIM with a tie on Citron. The scene-level comparisons complement the aggregate reconstruction results in Table 4.

4.5 Critique-to-Geometry Ablations

4.5.1 Extra Training versus Local Feedback

To validate the effectiveness of the Consistency Critique Agent, we compare localized refinement with additional baseline training. The baseline is retrained with the

Table 5 Comparison of Consistency Critique Agent (ours) with previous work on test images of the NeuMan dataset [12] using PSNR, SSIM and LPIPS metrics.

Method	Seattle			Citron			Parking		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Vid2Avatar	17.41	0.56	0.60	14.32	0.62	0.65	21.56	0.69	0.50
NeuMan	23.99	0.78	0.26	24.63	0.81	0.26	25.43	0.80	0.31
HUGS	26.48	0.85	0.10	24.35	0.80	0.12	26.96	0.84	0.13
Ours	28.57	0.93	0.05	24.85	0.81	0.10	27.94	0.90	0.09

Method	Bike			Jogging			Lab		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Vid2Avatar	14.86	0.51	0.69	15.04	0.41	0.70	13.96	0.60	0.68
NeuMan	25.55	0.83	0.23	22.70	0.68	0.32	24.96	0.86	0.21
HUGS	25.76	0.84	0.10	23.94	0.76	0.20	25.50	0.90	0.08
Ours	27.51	0.91	0.05	25.86	0.86	0.13	27.31	0.93	0.06

same number of extra optimization steps as the critique configuration. As shown in Table 6, extra training alone leads to only marginal PSNR improvement compared with the gain achieved by our agent. Both configurations start at 26.8 PSNR after 10k steps; another 5k steps increase the baseline to 27.3 and the critique configuration to 28.9.

Table 6 Effect of additional training steps. Extra training improves performance marginally compared to the gain from the Consistency Critique Agent. PSNR is reported.

Steps	Baseline	+ Critique
10k	26.8	26.8
10k + 5k	27.3	28.9

4.5.2 Reflection Rounds

We systematically evaluate the impact of reflection rounds on reconstruction performance using the NeuMan dataset. Table 7 shows substantial improvements over the initial reconstruction across all three metrics. The largest gain occurs after the first round. Round 3 achieves the highest PSNR and SSIM, while LPIPS is tied across Rounds 2–4; the small decreases in PSNR and SSIM at Round 4 indicate no further benefit from another round in this experiment. These results show that multi-round reflection effectively identifies and refines errors in fine-grained regions.

Table 7 Ablation study on the number of reflection rounds.

Setting	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$
Initial	0.842	26.723	0.115
Round 1	0.913	28.254	0.070
Round 2	0.914	28.348	0.068
Round 3	0.915	28.414	0.068
Round 4	0.914	28.386	0.068

4.5.3 Critic Model Scale

We evaluate the impact of the Consistency Critique Agent’s model scale on reconstruction quality. All variants are evaluated on NeuMan under the same CAL pipeline with three reflection rounds. Performance is evaluated using SSIM, PSNR, and LPIPS on test frames; we also record critique behavior and compute overhead under the A100/FP16 profiling setup.

Table 8 demonstrates a consistent trend: larger Qwen2.5-VL models lead to better reconstruction and perceptual metrics after refinement. Moving from 3B to 32B improves SSIM by 0.030 and PSNR by 0.90 dB, while reducing LPIPS by 0.023. The 32B model achieves the strongest results, while 7B offers a trade-off between quality and resources.

Table 9 shows the corresponding increase in latency. The 7B model requires 0.55 seconds per call, compared with 2.10 seconds for 32B. The average number of returned bad regions decreases from 5.8 for 3B to 2.1 for 32B.

4.6 Temporal Generation and Back-to-4D Ablations

Generation Agent and Back-to-4D Generative Distillation. Figure 6 and Table 10 present the Generation Agent ablation on MoVi. CFW, TC, LSFT, and ILP refer to Controllable Feature Warping, Temporal Convolution, Latent Space Fine-Tuning, and Iterative Latent Projection, respectively.

Table 8 Reconstruction and perceptual metrics when using different Qwen2.5-VL variants as the Consistency Critique Agent. All models are evaluated under the same CAL pipeline and three reflection rounds (see Table 7). Higher SSIM / PSNR and lower LPIPS are better.

Critique model	SSIM $\uparrow$	PSNR (dB) $\uparrow$	LPIPS $\downarrow$
No Critique	0.842	26.72	0.115
3B	0.884	27.45	0.092
7B	0.895	27.98	0.085
32B	0.914	28.35	0.069



Fig. 6 Qualitative ablation analysis. “w/o. LSFT” denotes results generated without Latent Space Fine-Tuning (LSFT) applied to the U-Net. Please see the supplementary video for further details.

Table 9 Behavioral and resource profiling of different Qwen2.5-VL variants. “Bad regions / frame” is the average number of bounding boxes returned per frame; “Latency” is average inference time per critique call (single A100, FP16). Critique is invoked every 8 frames (as in our main experiments).

Model	Bad regions	
	/ frame	Latency (s)
3B	5.8	0.20
7B	4.2	0.55
32B	2.1	2.10

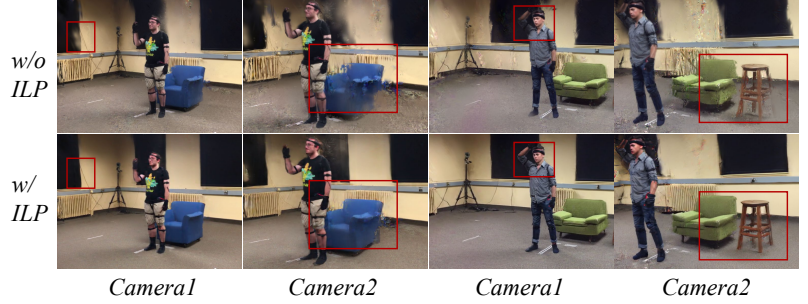


Fig. 7 Qualitative ablation studies demonstrate the impact of Back-to-4D generative distillation, where “w. ILP” denotes novel-camera results after the iterative Back-to-4D procedure.

Table 10 Ablation study on Generation Agent and Back-to-4D generative distillation (ILP).

Method	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$
Ours(w/o. CFW)	0.848	25.692	0.107
Ours(w/o. TC)	0.821	25.311	0.141
Ours(w/o. LSFT)	0.816	24.978	0.155
Ours(w/o. ILP)	0.855	26.103	0.102
Ours	0.873	26.335	0.092

The complete model improves all three metrics relative to each ablated configuration. Removing LSFT causes the largest reduction in SSIM and PSNR and the largest increase in LPIPS; temporal convolution also contributes substantially. Figure 6 demonstrates the effect of U-Net fine-tuning under large-scale motion, preserving human-body detail and improving the integration of the person with the scene.

As shown in Figure 7, Back-to-4D generative distillation, implemented through ILP, reduces artifacts when rendering from novel camera poses. The comparison illustrates the video diffusion model’s ability to refine illumination and human-scene appearance. Further implementation details are provided in Section 3.5.3.

4.7 Module Contributions

To understand the role of each agent in CAL, we remove one agent at a time while keeping the remaining compo-

nents intact (Table 11). Since the Reconstruction Agent provides CAL’s 3D backbone, removing it makes the end-to-end pipeline undefined; this setting is marked with dashes.

The remaining ablations show complementary roles. Removing Motion Guidance disables the motion conditioning pathway, including multi-scale condition features and their injection through SFT into the Generation Agent. Excluding the Generation Agent removes the diffusion model’s perceptual refinement, increasing FID-VID from 76.20 to 88.72. Omitting the Consistency Critique Agent leaves local geometric and human-scene artifacts uncorrected and produces the largest drop in PSNR and SSIM. All three removals lower T-Cons, demonstrating the contribution of these modules to the temporal quality of the final output.

4.8 Computational Cost

Table 12 profiles each agent at 512×512 output resolution. The profiling table uses one critique call every 8 frames; the source implementation description reports an interval of 8–12 frames. The time-aware encoder operates on VAE latents; its computation is reported as overlapping with Generation Agent denoising. Generation requires 0.45 seconds per frame, and the amortized full-pipeline entry is approximately 1.63 seconds

Table 11 End-to-end attribution study.

Configuration	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID-VID $\downarrow$	T-Cons $\uparrow$
w/o Reconstruction	-	-	-	-	-
w/o Motion Guidance	20.58	0.768	0.281	84.91	0.86
w/o Critique	19.93	0.748	0.298	88.45	0.84
w/o Generation	20.41	0.758	0.294	88.72	0.84
Full CAL (Ours)	21.93	0.802	0.223	76.20	0.92

Table 12 Compute profiling of each agent in the CAL framework at 512×512 output resolution. Memory indicates peak GPU usage during the corresponding module’s execution. Motion Guidance computation is reported as overlapping with Generation, and the Critique Agent is sparsely invoked. Reconstruction and critique are measured per sequence; generation and the amortized pipeline are reported per frame.

Stage / Agent	Peak GPU Mem.	Invocation	Runtime
Reconstruction Agent (3DGS)	2.1 GB	sequence-level optimization	0.17 h / sequence
Motion Guidance Agent (TAE)	2.5 GB	every frame	reported overlap with generation
Generation Agent (U-Net + VAE)	13.4 GB	every frame	0.45 s / frame
Consistency Critique Agent (32B)	-	1 / 8 frames, 2 rounds	0.16 h / sequence
Full CAL (amortized over long sequences)	14.2 GB	—	≈ 1.63 s / frame

Table 13 Time comparison and breakdown. Overall comparison (left) and breakdown of ours (right).

Methods	Time (h)	Stage	Time (h)
Neuman [12]	168	Reconstruction	0.17
HUGS [15]	0.5	Critique (2 rounds)	0.16
Ours	0.33	Animation	≈ 0 (real-time)
		Total	0.33

per frame. The amount of overlap depends on scheduling.

Table 13 compares geometry-preparation times and gives CAL’s stage breakdown. Initial reconstruction takes 0.17 hours and two critique rounds take 0.16 hours, totaling 0.33 hours. Direct Gaussian animation is real-time, with approximately zero cost on this hour scale. The total excludes diffusion-video synthesis. NeuMan and HUGS retain their benchmark timings of 168 and 0.5 hours; these values are reported under their respective experimental settings. Reconstruction and critique are measured per sequence, whereas generation is measured per frame; the full-pipeline timing is amortized over the sequence.

4.9 Novel Motion and Scene Applications

4.9.1 Supplied Novel Motion

Figure 8 shows results driven by AMASS motion-capture sequences [22], including running, jumping, and dancing, and by poses generated with MoMask [6] from text such as “jumping up” or “walking forward.” Across these motions, CAL produces coherent videos

while maintaining the subject’s appearance. A reference video of the target motion is not required.

4.9.2 Cross-Video Identity Consistency

Figure 9 demonstrates cross-video consistency during generation. Multiple videos of the same individual are generated with different pose guidance, while preserving the reference appearance. The comparisons with MIMO and AniGS illustrate CAL’s ability to maintain human identity and clothing detail across pose sequences.

4.9.3 Target-Scene Reenactment

As shown in Figure 10, the prepared human model is animated with predefined motion in the target scene, and the Generation Agent refines human–scene appearance for temporally coherent video reenactment.

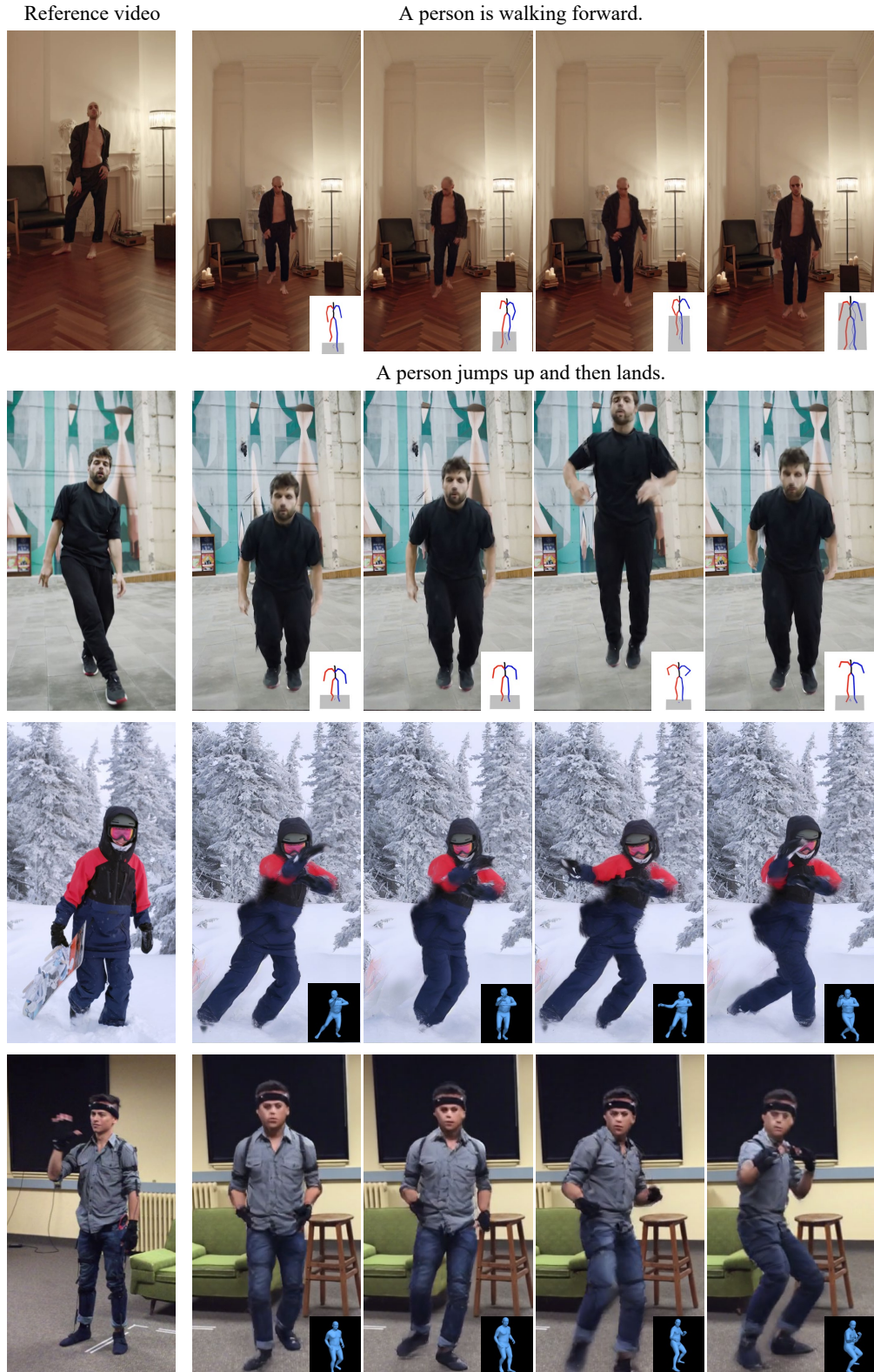


Fig. 8 Human animation with novel motion guidance. Given a reference video shown on the left, CAL animates the subject using motion sequences generated by MoMask from text or obtained from the AMASS dataset. The remaining columns show sampled frames from the generated sequences, with insets indicating the corresponding target poses. These examples illustrate motion control across different subjects and environments while retaining the subjects' appearance and clothing.

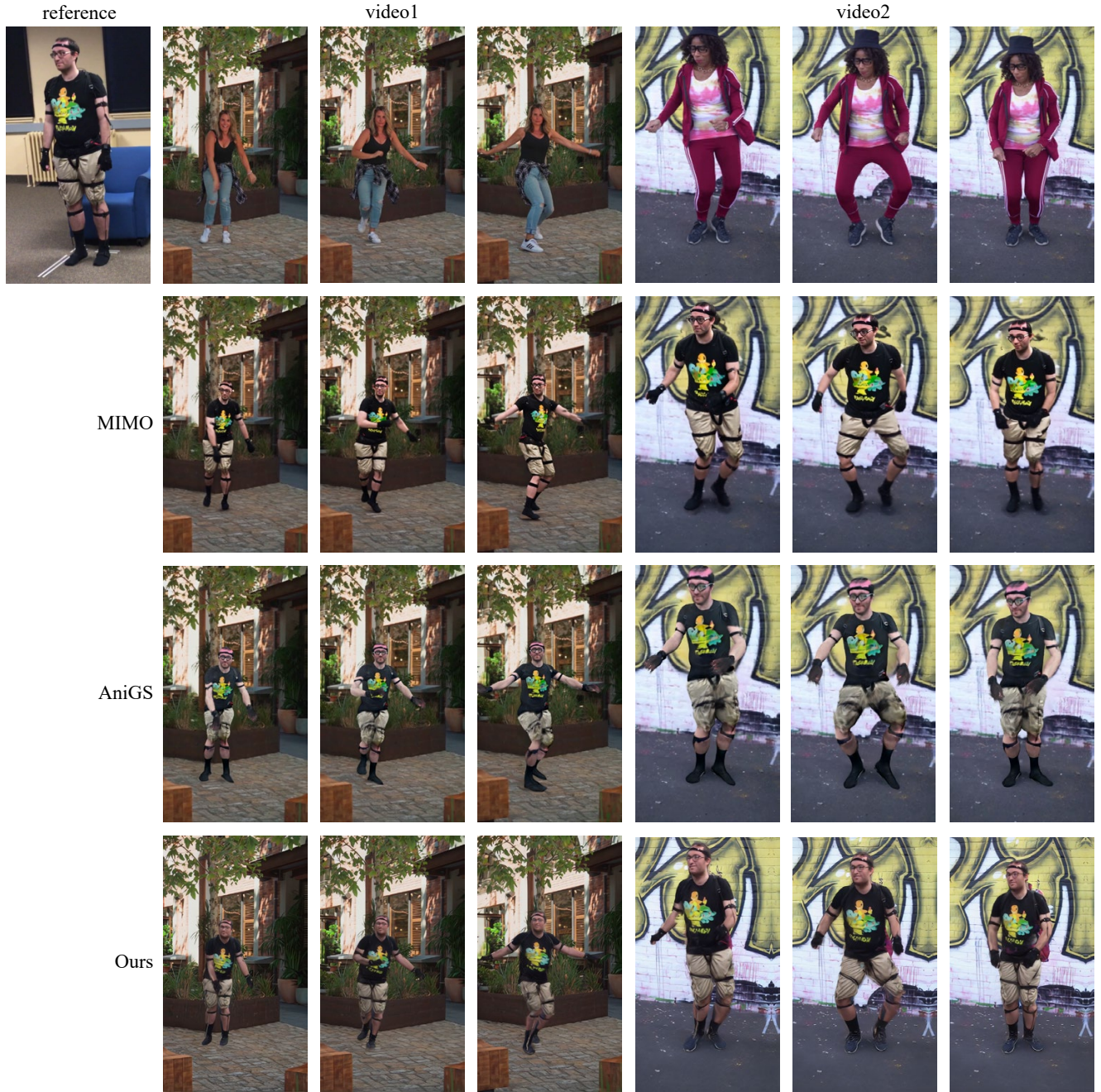


Fig. 9 Cross-video identity consistency under different motion guidance. The reference subject is shown at the upper left, and the top row presents two driving videos with different motions and backgrounds. The subsequent rows compare results from MIMO, AniGS, and CAL using the same reference identity. The comparison illustrates how each method preserves facial appearance, clothing details, and body proportions when animating the subject across the two driving sequences.

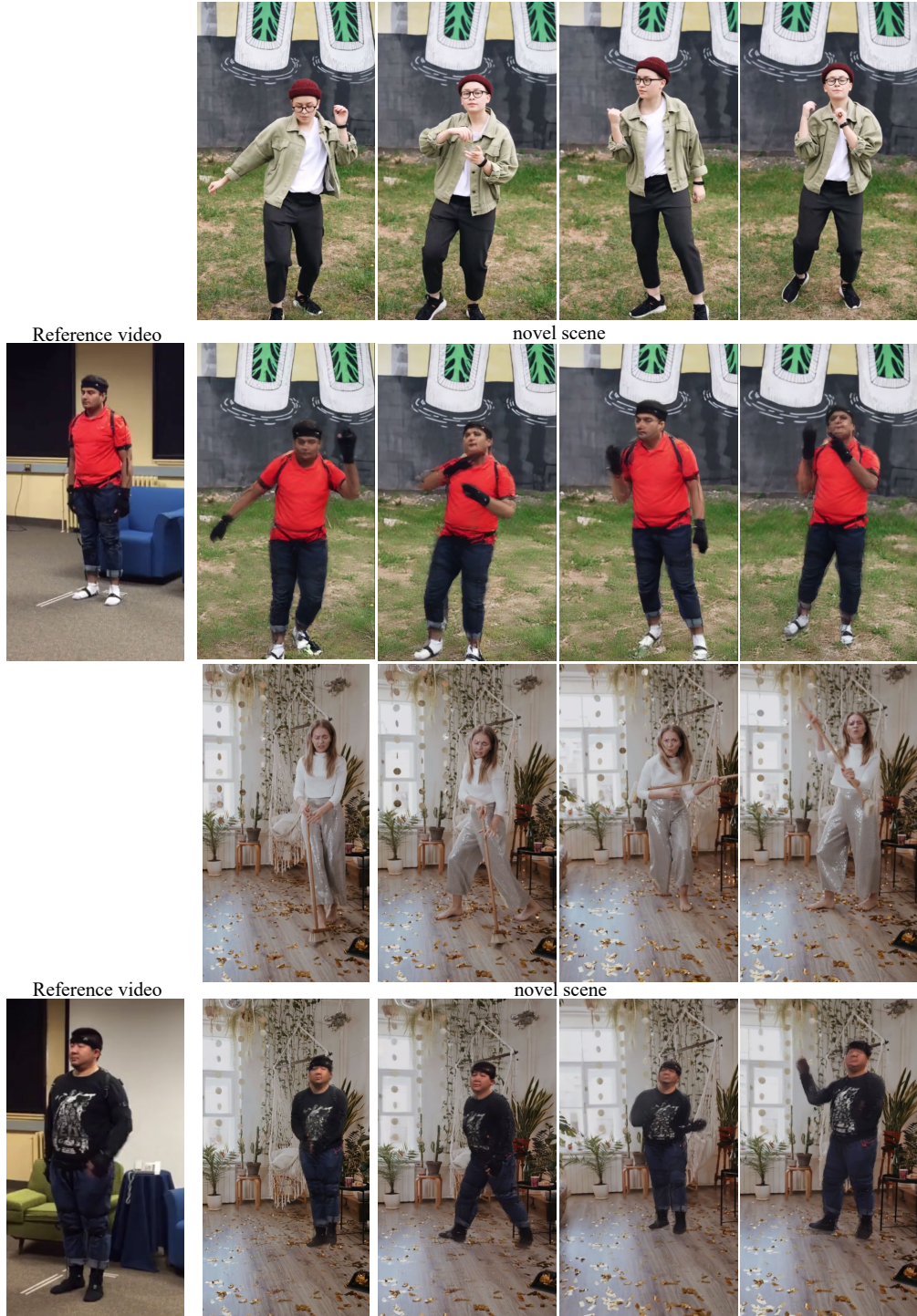


Fig. 10 Human reenactment in target scenes. For each example, the reference subject is shown on the left, while the upper and lower rows show the target video and CAL’s reenactment, respectively. CAL places the reference subject in the target environment and animates the subject according to the supplied pose sequence. The generated frames illustrate the preservation of the subject’s appearance and clothing while adapting the motion and human–scene composition to different environments.

5 Limitations

CAL remains sensitive to inaccurate SMPL estimates, particularly for facial expressions and fine-grained hand movements. Complex or dynamic scenes can also introduce reconstruction artifacts, since the current pipeline relies on COLMAP initialization and a static scene representation.

Monocular input provides limited supervision for unseen surfaces. Large pose or viewpoint changes may therefore expose incomplete geometry or uncertain appearance. Localized critique and generated novel-view supervision also depend on the reliability of the critic and video generator. Inaccurate feedback or hallucinated details can affect subsequent avatar refinement.

6 Impact Statement

Personalized human synthesis has applications in animation, VR/AR, and telepresence, but may also enable identity misuse or misleading video. Applications should obtain consent for source imagery and the identity being animated. Disclosure of synthesized content, digital watermarking, and provenance tracking can help mitigate these risks.

7 Conclusion

We introduced Collaborative Real2Sim-Sim2Real Agential Learning (CAL) to tightly couple geometric reconstruction and generative refinement for human dynamics. Within this framework, our Reconstruction Agent models human deformation through learned non-rigid offsets and residual skinning corrections, while the Consistency Critique Agent guides region-weighted 3DGS refinement using localized multimodal feedback. The Motion Guidance Agent applies supplied poses and encodes the resulting renderings as time-aware conditions, and the Generation Agent improves temporal coherence and visual realism through video diffusion. Back-to-4D uses generated novel-view sequences as fixed pseudo-targets for further Gaussian refinement. Together, our method supports expressive motion control, high-quality reconstruction, and coherent human-scene synthesis.

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A Implementation and Training Details

A.1 Parametric Body and Gaussian Rendering

The reconstruction combines SMPL [20] with 3D Gaussian Splatting [13]. SMPL maps identity-dependent shape β and articulated pose θ to a mesh. Its template with shape and pose blend shapes is

$$T(\beta, \theta) = \bar{T} + B_s(\beta) + B_p(\theta), \quad (17)$$

and the skinned mesh is

$$M(\beta, \theta) = W(T(\beta, \theta), J(\beta), \theta, \mathbf{W}_b). \quad (18)$$

Here J gives joint locations and $\mathbf{W}_b$ stores mesh skinning weights. CAL uses a canonical Da-pose mesh to initialize its human Gaussians and their LBS weights. Subject pose estimation supplies the captured articulation; new poses remain external conditions.

A Gaussian has center μ_k , scale $\mathbf{s}_k$, rotation $\mathbf{R}_k$, opacity o_k , and view-dependent appearance $\mathbf{c}_k$. Its covariance is

$$\Sigma_k = \mathbf{R}_k \text{diag}(\mathbf{s}_k^2) \mathbf{R}_k^\top. \quad (19)$$

This scale-and-rotation parameterization maintains a valid covariance. If $\tilde{G}_k(u)$ is its projected footprint at pixel u , the effective opacity and transmittance are

$$\alpha_k(u) = o_k \tilde{G}_k(u), \quad (20)$$

$$T_k(u) = \prod_{j < k} (1 - \alpha_j(u)). \quad (21)$$

Front-to-back compositing gives

$$C(u) = \sum_k T_k(u) \alpha_k(u) \mathbf{c}_k(\mathbf{v}(u)), \quad (22)$$

where $\mathbf{v}(u)$ is the viewing direction. Human and scene primitives participate in the same rendering operation after articulation and scene placement.

A.2 Reconstruction Networks and Losses

The spatial feature encoder follows triplane-style designs [3, 26]. It stores $\mathbf{F} \in \mathbb{R}^{3 \times h \times w \times d}$ and supplies a 64-dimensional local feature f^c for a queried canonical point. The non-rigid decoder $\mathcal{D}_{nr}$ is an MLP with three hidden layers of width 128. It predicts position, scale, and quaternion offsets. The rotation residual is composed as

$$\mathbf{q}_d = \mathbf{q}_c \otimes [1, \Delta q_1, \Delta q_2, \Delta q_3], \quad (23)$$

where $\otimes$ denotes the quaternion composition used by the implementation. Position and scale offsets follow Equations (1)–(3).

The LBS residual decoder $\mathcal{D}_r$ is a four-layer MLP of width 128 with ReLU and 24 joint-weight residual outputs. A two-layer color decoder predicts opacity and spherical-harmonic appearance from f^c . Human Gaussians use 50K sampled body-surface points and GauHuman adaptive control [10]. SMPL skinning weights are adjusted by $\mathcal{D}_r$, following HUGS [15].

Reconstruction networks train for 10k iterations on one NVIDIA A100, with a reported initial optimization time of about ten minutes. The stored Gaussian attributes and LBS weights enable direct animation after preparation. This time refers to the initial reconstruction stage; additional critique rounds and full diffusion-video generation are accounted for separately.

Reconstruction uses the image losses in Equation (6), with localized critique weighting as defined in Equation (9).

A.3 Conditioning and Temporal U-Net

The conditioning path and SFT modulation are defined in Equations (7) and (11). Table 14 specifies the spatial encoder and decoder stages, with temporal convolution and attention in the middle block.

The U-Net initialization uses Stable Diffusion V2.1. The temporal model trains for 30k iterations using Adam [14] at learning rate 10^{-3} , with eight A100-80G GPUs and batch size 48. Latent resolution is 64×64 with four channels, and the training sequence length is 6. The training procedure fixes pretrained spatial parameters and fine-tunes temporal layers.

A.4 Temporal Decoder and Sampling

The decoder trains on latent sequences sampled by the fine-tuned U-Net with corresponding ground-truth videos. Its schedule is 10k iterations on MoVi followed by 15k on HumanVid, learning rate 5×10^{-5} , batch size 8, and length-5 sequences. Temporal residual processing and CFW are incorporated; pretrained spatial layers are held fixed according to the recorded procedure. Training uses the hybrid objective in Equation (12).

The Generation Agent uses motion-guided diffusion sampling [39], with 50 sampling steps per sequence and progressive patch aggregation for arbitrary resolutions [33].

B Evaluation Protocol Details

B.1 User Study Questionnaire

We developed a web-based interface for the user study. For each query, participants were shown a reference video and generated results from CAL and the baseline methods (AnimateAnyone, AniGS, MIMO, and Champ) in randomized order, with method identities hidden. Participants rated the videos on a Likert scale from 1 to 5 using the three criteria in Table 15.

The 25 participants consisted of 10 computer-vision researchers and 15 non-experts to balance technical scrutiny with general perceptual preference. Each participant evaluated 15 randomly sampled sequences, resulting in 375 participant-sequence evaluations per method. Table 3 reports the averaged User score.

Table 14 Simplified architecture of the Generation Agent’s U-Net. The model processes frames spatially in the Encoder and Decoder, with **temporal mixing confined exclusively to the Middle Block**. Sequence length is 6 for U-Net training and 5 for decoder training.

Stage	Resolution	Channels	Key Operations
Input	64×64	4	Latent Input
Encoder (Spatial Only)	$64 \rightarrow 32$ $32 \rightarrow 16$ $16 \rightarrow 8$ 8×8	$320 \rightarrow 640$ $640 \rightarrow 1280$ $1280 \rightarrow 1280$ 1280	Spatial ResBlock + Attn, Downsample Spatial ResBlock + Attn, Downsample Spatial ResBlock + Attn, Downsample Spatial ResBlock (No Attn)
MidBlock (Spatial + Temporal)	8×8	1280	1. Spatial ResBlock + Attn 2. Temporal Convolution (3D) 3. Temporal Attention
Decoder (Spatial Only)	8×8 $8 \rightarrow 16$ $16 \rightarrow 32$ $32 \rightarrow 64$	1280 $1280 \rightarrow 1280$ $1280 \rightarrow 640$ $640 \rightarrow 320$	Spatial ResBlock (No Attn) Spatial ResBlock + Attn, Upsample Spatial ResBlock + Attn, Upsample Spatial ResBlock + Attn, Upsample
Output	64×64	4	Norm $\rightarrow$ SiLU $\rightarrow$ Conv2D (3×3)

Table 15 Questionnaire and criteria provided to the 25 participants during the user study.

Criterion	Question / Instruction
Motion Fidelity	“How accurately does the character follow the reference motion? Ignore background artifacts and focus on limb articulation.” (1: Completely Wrong, 5: Perfect Alignment)
Identity Preservation	“Does the character in the video look like the person in the reference image? Look at facial features and clothing details.” (1: Different Person, 5: Same Person)
Visual Realism	“How realistic is the overall video? Consider lighting consistency, shadow correctness, and temporal stability (flickering).” (1: Artificial/Glitchy, 5: Photorealistic)

C Full Critique Prompts

Figures 11 and 12 illustrate the full textual prompts used in our Consistency Critique Agent. These prompts guide the MLLM in performing two tasks: Robust Reconstruction Filtering and Spatial Fine-Grained Reconstruction. The filtering prompt requests a reconstruction score and analysis, while the spatial prompt requests localized regions and feedback. These responses guide input screening and focused refinement in the reconstruction pipeline.

Prompt: Critique Agent — Robust Reconstruction Filtering

You are an expert system for evaluating the quality of SMPL-based 3D human reconstructions.

Your task is to compare an original image with a rendered image of a 3D digital human. This rendered image was created using parameters derived from SMPL pose estimation.

From 0 to 100, assign a score to the rendered image, where 0 means the SMPL estimation is completely inaccurate and 100 means it matches the person in the original image almost perfectly.

Carefully analyze the alignment between the 3D human and the person in the original image, focusing on:

- Body pose and limb positions
- Silhouette consistency and proportions
- Major mismatches or distortions

Provide your final answer strictly in the following JSON format:

```
'''json
{"Final Score": score_value, "Estimate Analysis": analysis_string}
'''
```

- score_value: an integer from 0–100 representing how well the SMPL-based reconstruction reflects the original image.
- analysis_string: a concise explanation for the given score, highlighting key reasons for accuracy or error.

Do not include any code in your answer.

Fig. 11 Prompt for Robust Reconstruction Filtering in our Consistency Critique Agent.

System prompt: Critique Agent — Spatial Fine-Grained Reconstruction

When reconstructing 3D human models from monocular videos using 3D Gaussian Splatting (3DGS), challenges can arise, leading to poorly reconstructed human body regions. To accurately identify these imperfections in rendered images, it's essential to be aware of these common failure modes:

- **Blurry or Distortion:** Pay close attention to areas such as the face, hands, and feet. These might lack clear definition, appearing blurry, distorted, or not sharply rendered, with their natural shape compromised.
- **Intersections:** A key indicator of poor reconstruction is the presence of unnatural overlaps. Specifically, watch for instances where hands or arms unnaturally intersect with other body elements or clothing, such as pockets, the torso, or the head, severely compromising the model's realism.
- **Ground Fusion:** Observe any unclear separation between the feet and the floor. They may appear partially merged with the ground plane, giving a "smudged" or "melted" appearance rather than standing distinctly.
- **Artifacts:** These often appear as ellipsoidal artifacts produced by 3DGS, especially noticeable around the edges of the human torso or clothes.

Understanding these concepts enables you to effectively identify poorly reconstructed human body regions in rendered images.

Prompt: Critique Agent — Spatial Fine-Grained Reconstruction

You are an expert system for assessing the quality of 3DGS human reconstructions.

Your **task** is to compare an original image with a rendered image to identify and locate poorly reconstructed human body regions.

First, given an original image and a rendered image, locate human's right and left hand, right and left foot, and face in the rendered image. Use the bounding box coordinates to locate these regions.

Second, analyze and identify the possible artifacts, blurry or distortion, intersections, and ground fusion in the above five regions (by comparing with the original image). For each region, provide a descriptive label.

You **MUST** identify and label these human body regions in the 3DGS rendered image. For each human body region, provide its **bounding box coordinates** from the rendered image and a **descriptive label**. If a human body region is well-reconstructed, explicitly label it as "well-reconstructed [body part]". Do not include any code.

Your final output must be in JSON format like this:

```
'''json
[
  {
    "id": "right hand",
    "bbox_2d": [x_min, y_min, x_max, y_max],
    "label": "descriptive label"
  },
  {
    "id": "left hand",
    "bbox_2d": [x_min, y_min, x_max, y_max],
    "label": "descriptive label"
  },
  ...
]
```

Output the bounding box coordinates and a descriptive label for each identified area using the specified *JSON format*.

Fig. 12 Prompt for Spatial Fine-Grained Reconstruction in our Consistency Critique Agent.